

Lab 05

Due March 3, 5:00 PM PST

AUTHOR

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Problem 1a

write your answer here

```
knitr::opts_knit$set(root.dir = "/home/stabby/stat-240/RWorkspace/stat240/labs/lab_05")
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

```
✓ dplyr      1.1.4      ✓ readr      2.1.6
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.1      ✓ tibble     3.3.1
✓ lubridate  1.9.4      ✓ tidyr      1.3.2
✓ purrr      1.2.1
```

— Conflicts — tidyverse_conflicts() —

```
* dplyr::filter() masks stats::filter()
```

```
* dplyr::lag()     masks stats::lag()
```

i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```
ebird <- read.csv("ebird30.csv")
names <- read.csv("names.csv")

source("canada.R")
```

Loading required package: viridisLite

Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE

Registered S3 method overwritten by 'geojsonsf':

```
method      from
print.geojson geojson
```

Attaching package: 'geojsonio'

The following object is masked from 'package:base':

```
pretty
```

```
one_region <- which(apply(ebird[, 2:14], 1, function(x) sum(x > 0)) == 1)

ebird[one_region[1], ]
```

	Code	CA.AB	CA.BC	CA.MB	CA.NB	CA.NL	CA.NT	CA.NS	CA.NU	CA.ON	CA.PE	CA.QC
7	rosgoo	0	0	0	0	0	0	0	0	5	0	0
	CA.SK	CA.YT										
7	0	0										

```
ebird[one_region[2], ]
```

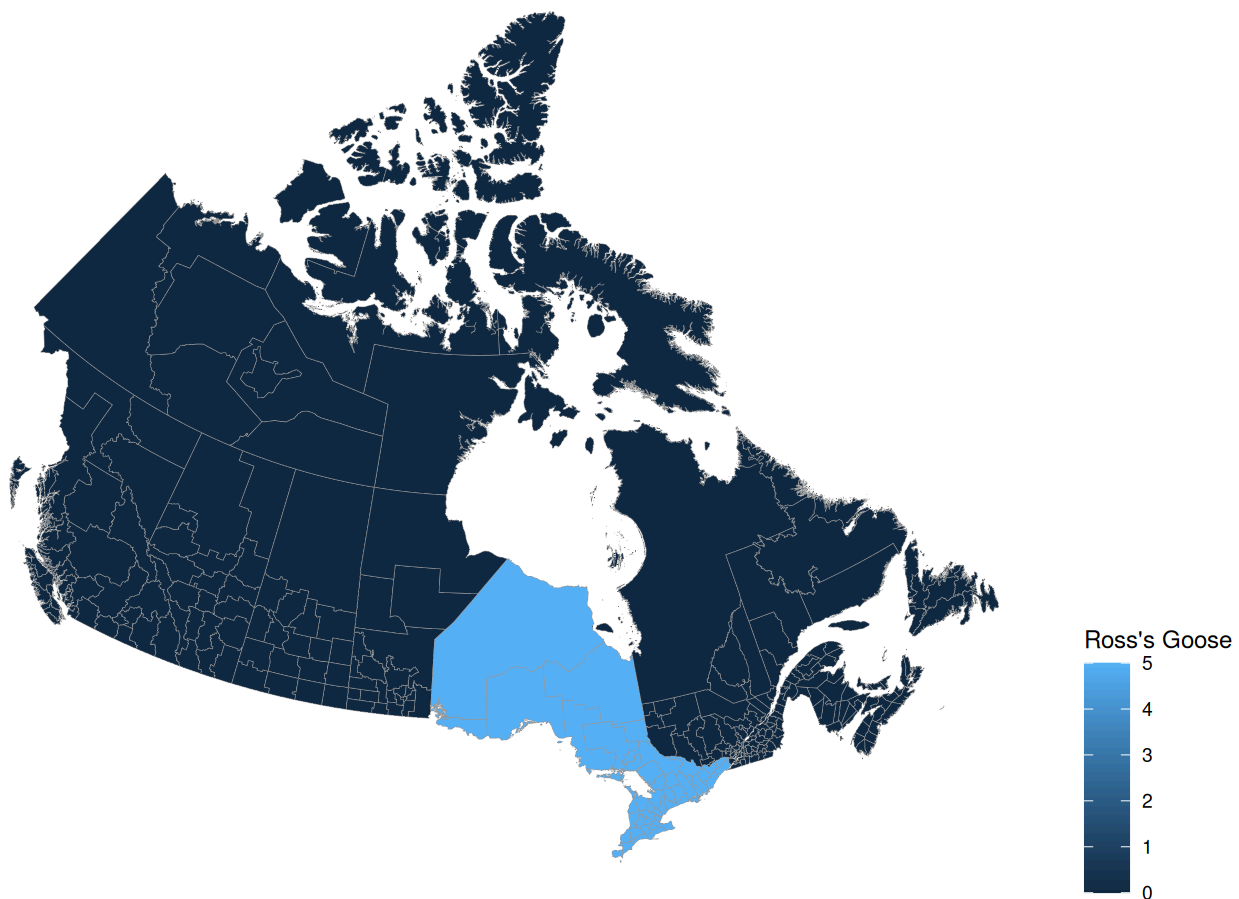
	Code	CA.AB	CA.BC	CA.MB	CA.NB	CA.NL	CA.NT	CA.NS	CA.NU	CA.ON	CA.PE	CA.QC
9	gragoo	0	0	0	0	0	0	0	0	4	0	0
	CA.SK	CA.YT										
9	0	0										

```
one_name <- names[one_region[1], ]
comm_name <- one_name$Common.Name
heat <- list()

for(i in 1:13 ){
  heat[[colnames(ebird)[i+1]]] = ebird[one_region[1], colnames(ebird)[i+1]]
}
canada(title = comm_name, heat = heat)
```

Loading required package: grid

Warning: The `size` argument of `element\_rect()` is deprecated as of ggplot2 3.4.0.
i Please use the `linewidth` argument instead.



Problem 1b

write answer here

```
row_sum <- numeric(nrow(ebird))

for (i in 1:nrow(ebird)) {
  for (j in 2:ncol(ebird)) {
    row_sum[i] <- row_sum[i] + ebird[i, j]
  }
}
output <- data.frame(Common_Name = names$Common.Name, Scientific_Name = names$Scientific.
finalb <- output |> arrange(desc(Sum))
head(finalb, n= 1)
```

	Common_Name	Scientific_Name	Sum
1	Canada Goose	Branta canadensis	123096

Problem 1c

write answer here

```
newFin <- finalb |> filter(Sum != 0) |> arrange(Sum)
minimum <- min(newFin$Sum)
minName <- newFin |> filter(Sum == minimum)
finName <- sort(c(minName$Common_Name))
head(finName, n= 10)
```

```
[1] "American Wigeon x Mallard (hybrid)" "Blue-winged Teal"
[3] "Broad-winged Hawk"                  "Clay-colored Sparrow"
[5] "Common Yellowthroat"                "Dickcissel"
[7] "Dusky Grouse"                       "Golden Pheasant"
[9] "Graylag x Swan Goose (hybrid)"      "Great Egret"
```